Irish Trade Analysis

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Setup chunk (libraries + options)

```
library(tidyverse)
library(lubridate)
library(stringr)
```

Phase 1:

Load & Clean Data

```
trade_raw <- read_csv("raw.csv")
#glimpse(trade_raw)
```

At this point, our columns are:

- `Statistic Label`
- `Month`
- `Commodity Group`
- `UNIT`
- `VALUE`

We'll **never modify `trade_raw` directly** again.

Clean and standardise variables

```
trade_clean <- trade_raw %>%
  rename(
    statistic = `Statistic Label`,
    month_raw = Month,
    commodity_raw = `Commodity Group`,
    unit = UNIT,
    value = VALUE
  ) %>%
  mutate(
    trade_type = case_when(
      str_detect(statistic, "Imports") ~ "Import",
      str_detect(statistic, "Exports") ~ "Export"
    ),
    month = as.Date(paste0("01 ", month_raw), format = "%d %Y %B"),
    year = lubridate::year(month),
    commodity_code = str_extract(commodity_raw, "\\(([^)]+)\\)") %>%
      str_remove_all("[()]"),
    commodity_desc = str_remove(commodity_raw, "\\s*\\([^)]*\\)"),
    hierarchy_level = if_else(
      nchar(commodity_code) == 1,
      "top",
      "detail"
    )
  ) %>%
  select(
    trade_type,
    month,
    year,
    commodity_code,
    commodity_desc,
```

```
    hierarchy_level,
    value
  )
```

What this restructuring block does (important)

- Converts `"1972 January"` → real `Date`
- Separates **Import vs Export**
- Extracts: - `commodity_code` → `0`, `01`, `54`, etc.
- `commodity_desc` → clean names
- Tags hierarchy explicitly (`top` vs `detail`)

This is **correct handling** of SITC-style data.

Split datasets (no double counting ever)

```
trade_top_level <- trade_clean %>%
  filter(hierarchy_level == "top")

trade_detail_level <- trade_clean %>%
  filter(hierarchy_level == "detail")

n_distinct(trade_top_level$commodity_code)
```

```
[1] 10
```

```
n_distinct(trade_detail_level$commodity_code)
```

```
[1] 64
```

From now on:

- Macro analysis → `trade_top_level`
- Sector deep dives → `trade_detail_level`

We will **never mix them**.

Quick sanity checks

```
range(trade_clean$year)
```

```
[1] 1972 2025
```

```
unique(trade_clean$trade_type)
```

```
[1] "Import" "Export"
```

```
anyNA(trade_clean$month)
```

```
[1] FALSE
```

We could see:

- Years from **1972 to 2025**
- Two trade types only
- No missing dates

Where we are now

At this point, we have:

- Clean, reproducible data
- Correct hierarchy handling
- A strong analytical base

This alone is already **better than 80% of public trade analyses**.

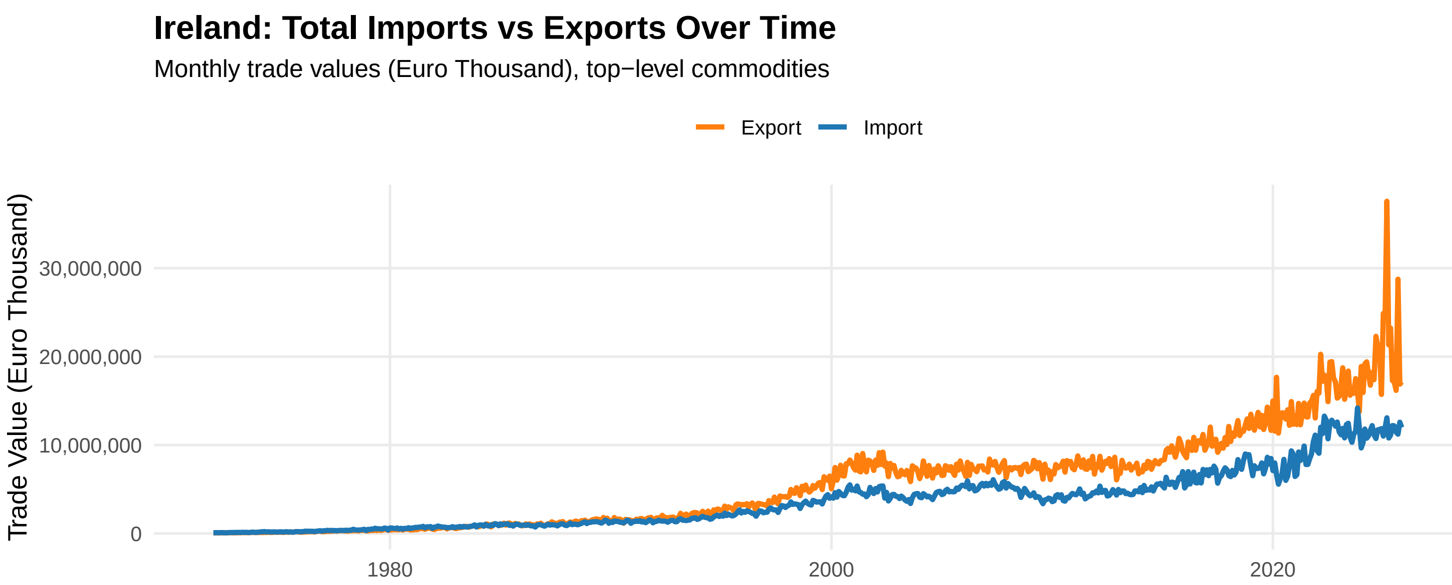
Phase 2:

2.1 Aggregate total imports & exports over time

```
trade_totals <- trade_top_level %>%
  group_by(month, trade_type) %>%
  summarise(
    total_value = sum(value, na.rm = TRUE),
    .groups = "drop"
  )
```

2.2 Imports vs exports plot

```
ggplot(trade_totals, aes(x = month, y = total_value, colour = trade_type)) +
  geom_line(linewidth = 1.2) +
  scale_colour_manual(
    values = c("Import" = "#1F77B4", "Export" = "#FF7F0E")
  ) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Ireland: Total Imports vs Exports Over Time",
    subtitle = "Monthly trade values (Euro Thousand), top-level commodities",
    x = NULL,
    y = "Trade Value (Euro Thousand)",
    colour = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    legend.position = "top",
    panel.grid.minor = element_blank()
  )
```

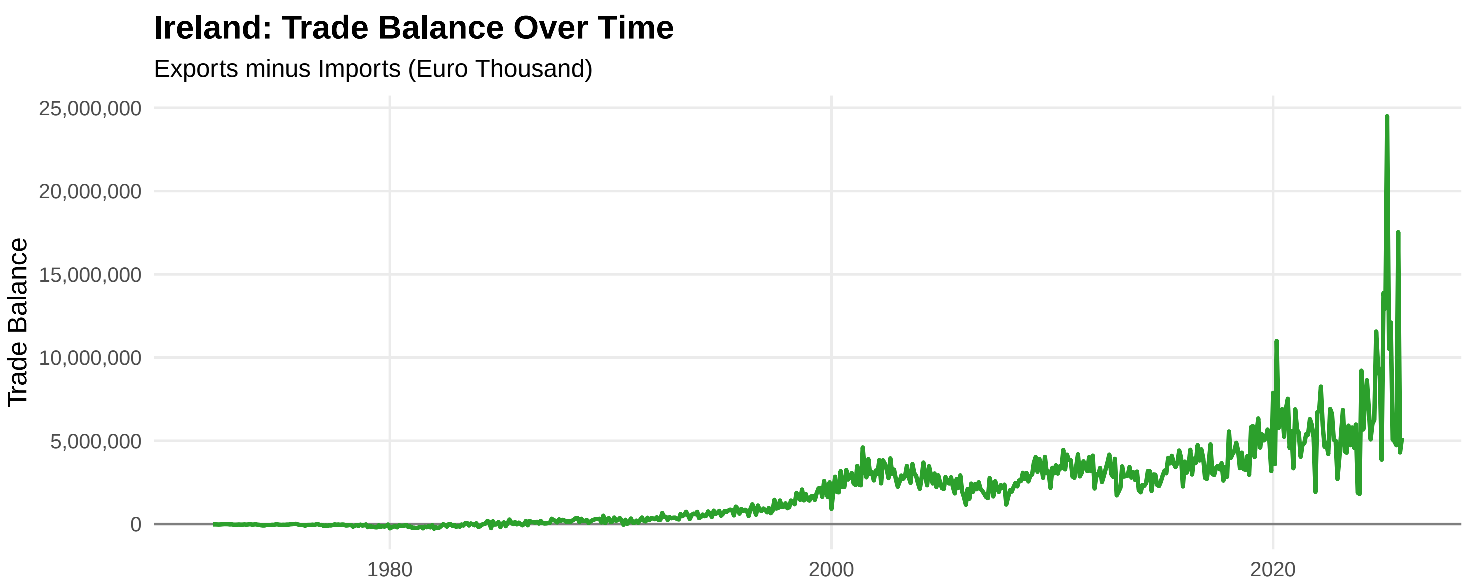


2.3 Trade balance calculation (Exports − Imports)

```
trade_balance <- trade_totals %>%
  pivot_wider(
    names_from = trade_type,
    values_from = total_value
  ) %>%
  mutate(
    trade_balance = Export - Import
  )
```

2.4 Trade balance plot

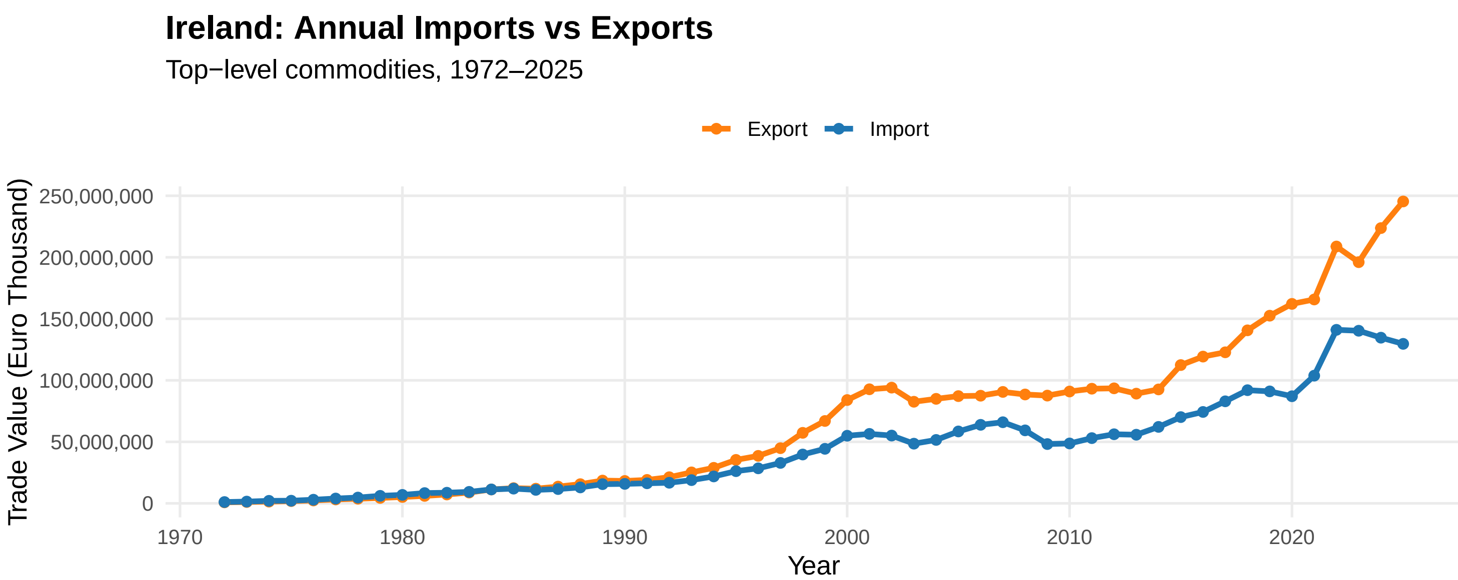
```
ggplot(trade_balance, aes(x = month, y = trade_balance)) +
  geom_hline(yintercept = 0, colour = "grey50", linewidth = 0.6) +
  geom_line(colour = "#2CA02C", linewidth = 1) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Ireland: Trade Balance Over Time",
    subtitle = "Exports minus Imports (Euro Thousand)",
    x = NULL,
    y = "Trade Balance"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12),
    panel.grid.minor = element_blank()
  )
```



2.5 Structural trend (annual smoothing)

```
trade_annual <- trade_top_level %>%
  group_by(year, trade_type) %>%
  summarise(
    total_value = sum(value, na.rm = TRUE),
    .groups = "drop"
  )
```

```
ggplot(trade_annual, aes(x = year, y = total_value, colour = trade_type)) +
  geom_line(linewidth = 1.3) +
  geom_point(size = 2) +
  scale_colour_manual(
    values = c("Import" = "#1F77B4", "Export" = "#FF7F0E")
  ) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Ireland: Annual Imports vs Exports",
    subtitle = "Top-level commodities, 1972-2025",
    x = "Year",
    y = "Trade Value (Euro Thousand)",
    colour = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    legend.position = "top",
    panel.grid.minor = element_blank()
  )
```

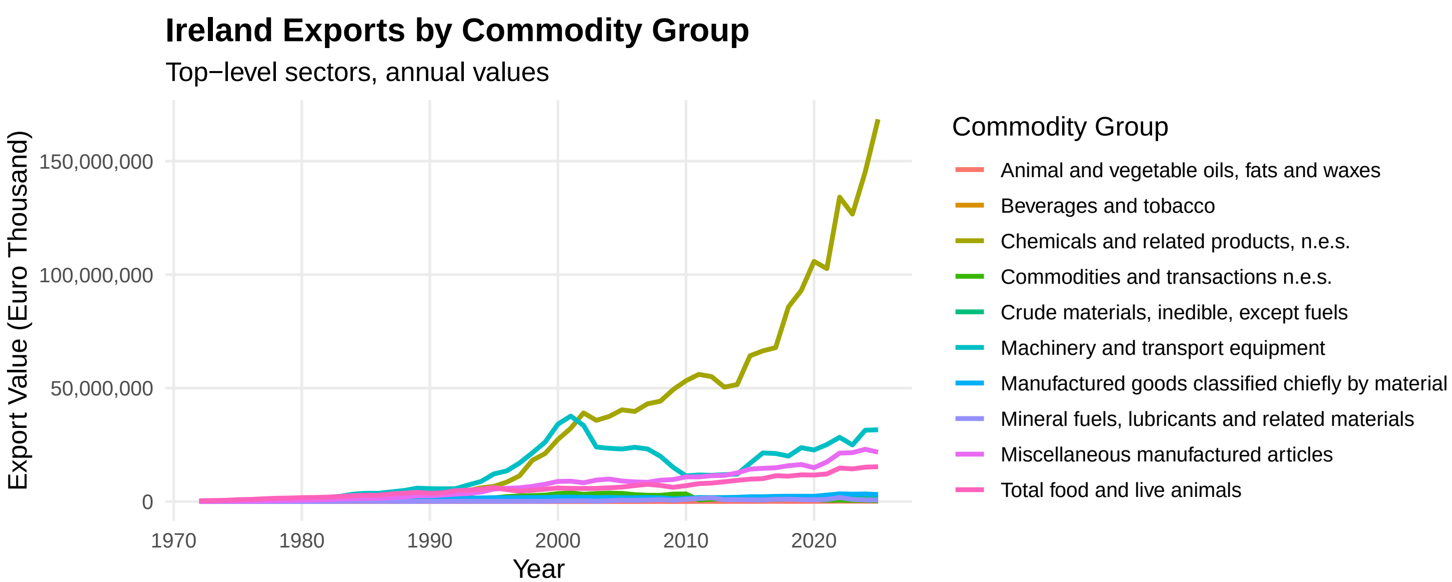

Ireland: Annual Imports vs Exports
Top−level commodities, 1972–2025
Export
Import
250,000,000
200,000,000
150,000,000
100,000,000
50,000,000
0
Trade Value (Euro Thousand)
1970
1980
1990
2000
2010
2020
Year

Phase 3:

3.1 Top export commodity groups over time

```
exports_by_sector <- trade_top_level %>%
  filter(trade_type == "Export") %>%
  group_by(year, commodity_desc) %>%
  summarise(
    export_value = sum(value, na.rm = TRUE),
    .groups = "drop"
  )
```

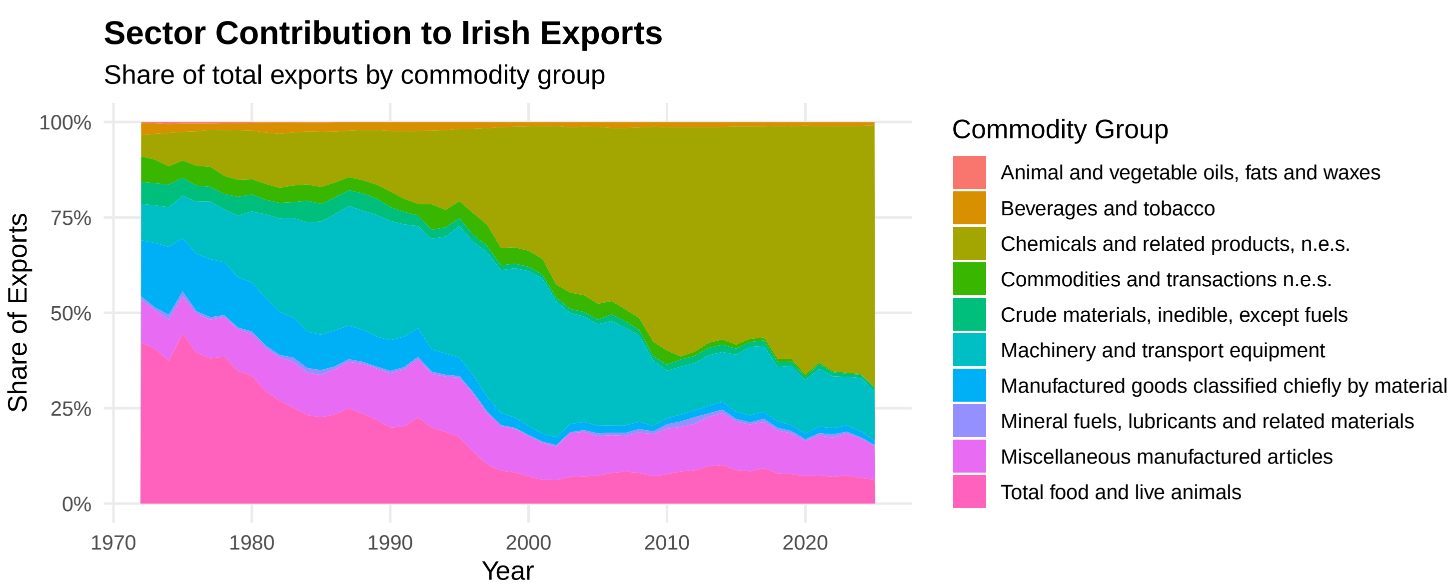
```
ggplot(exports_by_sector,
       aes(x = year, y = export_value, colour = commodity_desc)) +
  geom_line(linewidth = 1.1) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Ireland Exports by Commodity Group",
    subtitle = "Top-level sectors, annual values",
    x = "Year",
    y = "Export Value (Euro Thousand)",
    colour = "Commodity Group"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    legend.position = "right",
    panel.grid.minor = element_blank()
  )
```



3.2 Sector contribution share

```
export_shares <- exports_by_sector %>%
  group_by(year) %>%
  mutate(
    total_exports = sum(export_value),
    share = export_value / total_exports
  ) %>%
  ungroup()
```

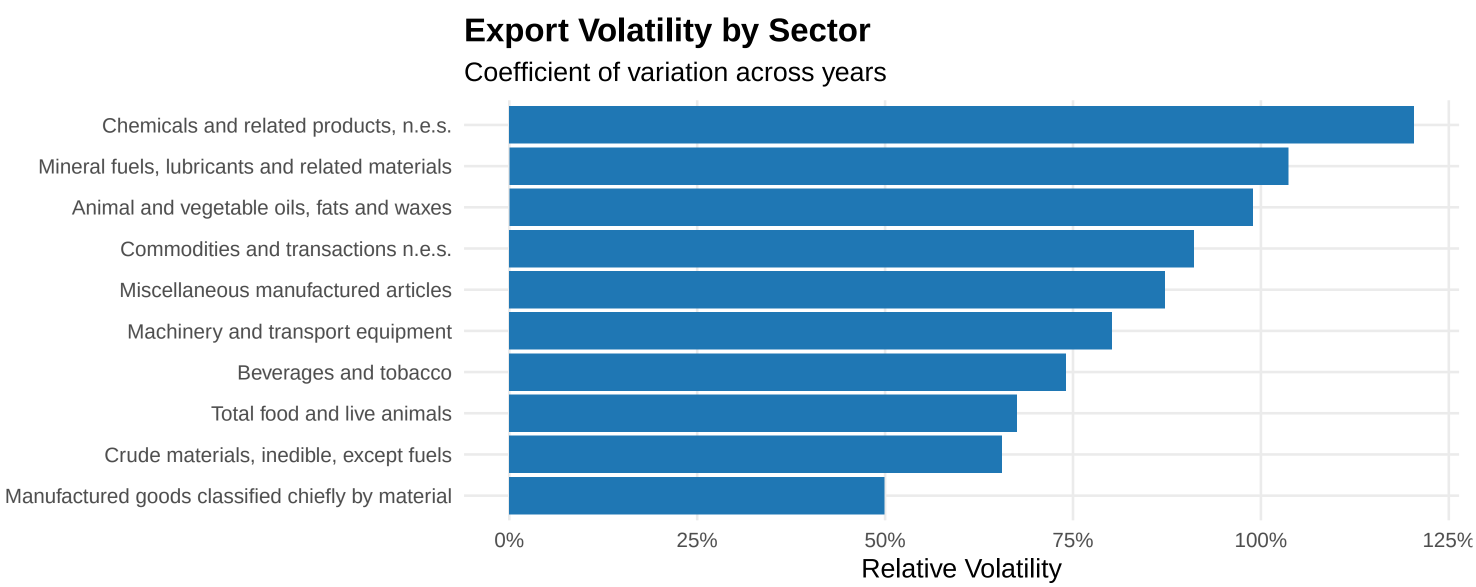
```
ggplot(export_shares,
       aes(x = year, y = share, fill = commodity_desc)) +
  geom_area() +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Sector Contribution to Irish Exports",
    subtitle = "Share of total exports by commodity group",
    x = "Year",
    y = "Share of Exports",
    fill = "Commodity Group"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank(),
    legend.position = "right"
  )
```



3.3 Volatility comparison across sectors

```
export_volatility <- exports_by_sector %>%
  group_by(commodity_desc) %>%
  summarise(
    mean_exports = mean(export_value),
    sd_exports = sd(export_value),
    cv = sd_exports / mean_exports,
    .groups = "drop"
  )
```

```r
ggplot(export_volatility,
       aes(x = reorder(commodity_desc, cv), y = cv)) +
  geom_col(fill = "#1F77B4") +
  coord_flip() +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Export Volatility by Sector",
    subtitle = "Coefficient of variation across years",
    x = NULL,
    y = "Relative Volatility"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank()
  )
```

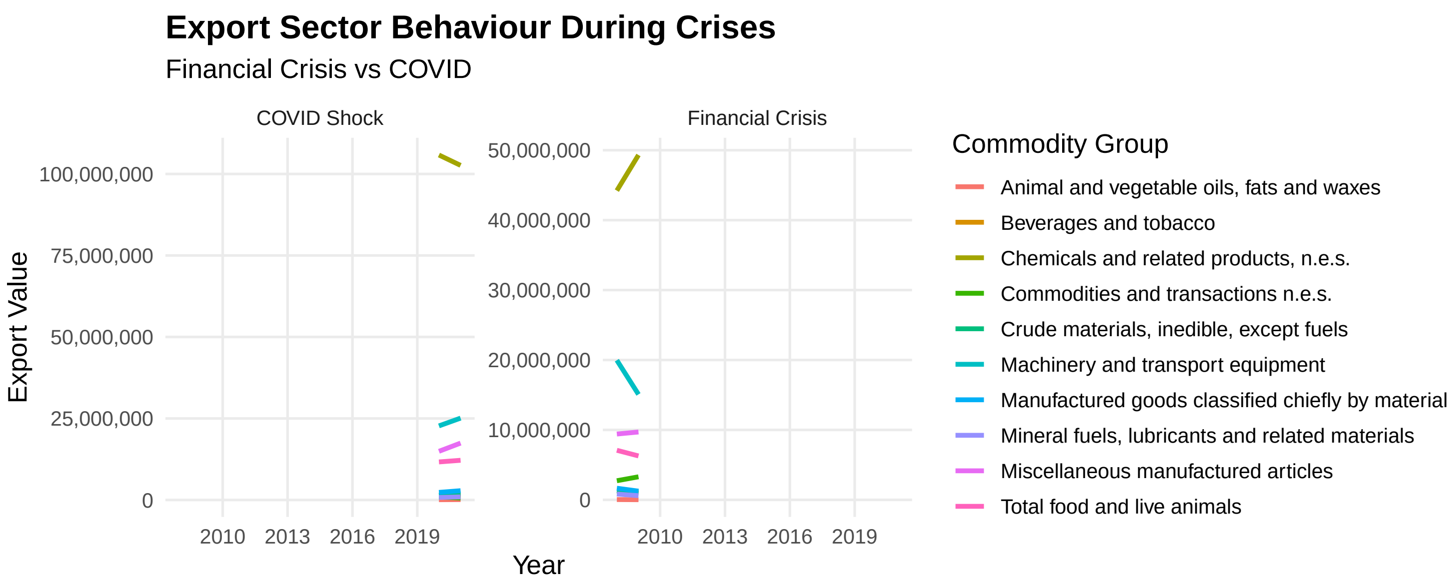


3.4 Crisis Analysis (2008 vs COVID)

```r
exports_crisis <- exports_by_sector %>%
  mutate(
    period = case_when(
      year %in% 2008:2009 ~ "Financial Crisis",
      year %in% 2020:2021 ~ "COVID Shock",
      TRUE ~ "Normal"
    )
  ) %>%
  filter(period != "Normal")
```

```r
ggplot(exports_crisis,
       aes(x = year, y = export_value, colour = commodity_desc)) +
  geom_line(linewidth = 1.1) +
  facet_wrap(~ period, scales = "free_y") +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Export Sector Behaviour During Crises",
    subtitle = "Financial Crisis vs COVID",
    x = "Year",
    y = "Export Value",
```

```
    colour = "Commodity Group"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank(),
    legend.position = "right"
  )
```



Phase 4:

4.1 Regime Analysis

Let’s define economic regimes:

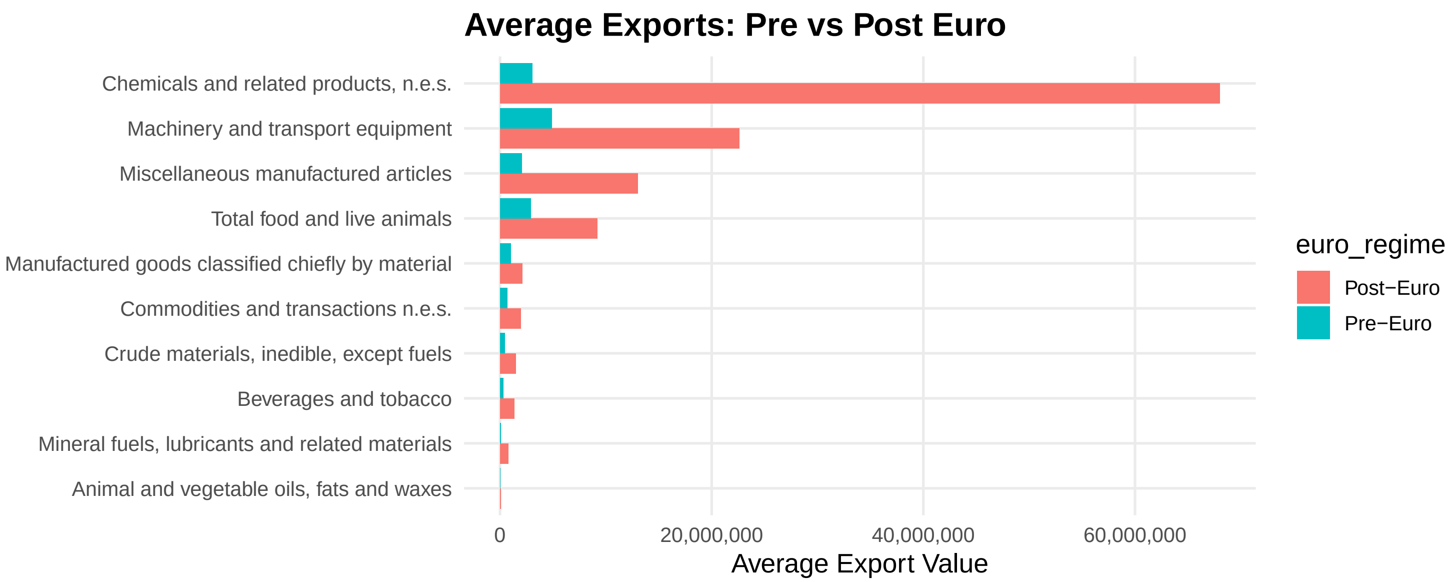
- Pre-Euro: before 1999
- Post-Euro: 1999 onward
- Pre-Brexit: before 2016
- Post-Brexit: 2016 onward

```r
exports_regime <- exports_by_sector %>%
  mutate(
    euro_regime = if_else(year < 1999, "Pre-Euro", "Post-Euro"),
    brexit_regime = if_else(year < 2016, "Pre-Brexit", "Post-Brexit")
  )
```

Compare average exports

```r
regime_summary <- exports_regime %>%
  group_by(euro_regime, commodity_desc) %>%
  summarise(
    avg_exports = mean(export_value),
    .groups = "drop"
  )
```

```r
ggplot(regime_summary,
       aes(x = reorder(commodity_desc, avg_exports),
           y = avg_exports,
           fill = euro_regime)) +
  geom_col(position = "dodge") +
  coord_flip() +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Average Exports: Pre vs Post Euro",
    x = NULL,
    y = "Average Export Value"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank()
  )
```



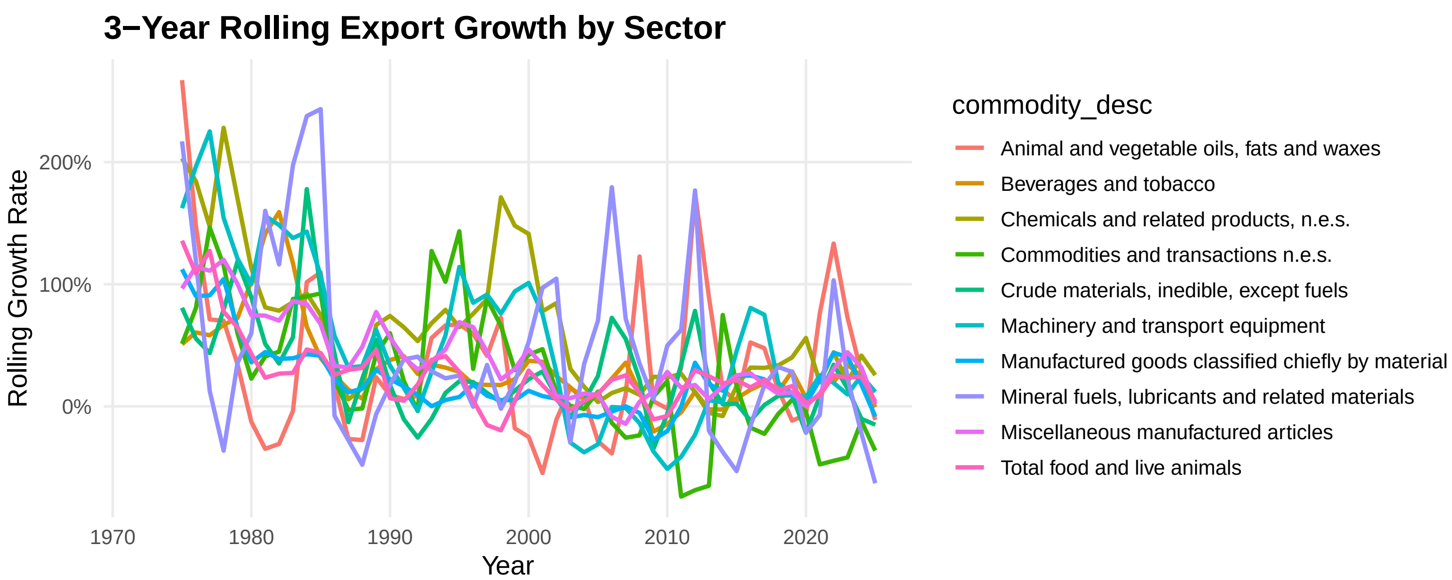
4.2 Rolling Growth Rates

We compute 3-year rolling growth

```
library(slider)

exports_growth <- exports_by_sector %>%
  arrange(commodity_desc, year) %>%
  group_by(commodity_desc) %>%
  mutate(
    rolling_growth = (export_value /
      lag(export_value, 3)) - 1
  ) %>%
  ungroup()
```

```
ggplot(exports_growth,
       aes(x = year, y = rolling_growth,
           colour = commodity_desc)) +
  geom_line(linewidth = 1) +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "3-Year Rolling Export Growth by Sector",
    x = "Year",
    y = "Rolling Growth Rate"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank(),
    legend.position = "right"
  )
```



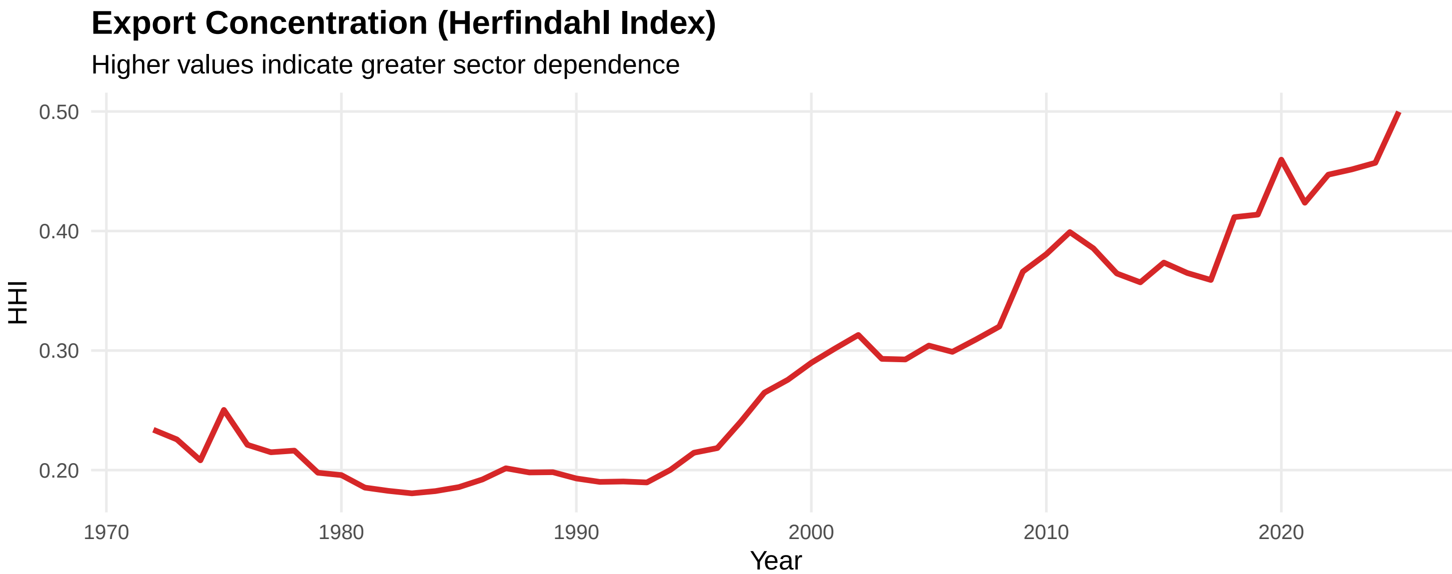
4.3 Concentration Risk (Herfindahl Index)

Now we quantify dependence formally.

HHI = sum of squared sector shares.

```
hhi_data <- export_shares %>%
  group_by(year) %>%
  summarise(
    hhi = sum(share^2),
    .groups = "drop"
  )
```

```
ggplot(hhi_data, aes(x = year, y = hhi)) +
  geom_line(colour = "#D62728", linewidth = 1.3) +
  scale_y_continuous(labels = scales::number_format(accuracy = 0.01)) +
  labs(
    title = "Export Concentration (Herfindahl Index)",
    subtitle = "Higher values indicate greater sector dependence",
    x = "Year",
    y = "HHI"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    panel.grid.minor = element_blank()
  )
```


Export Concentration (Herfindahl Index)
Higher values indicate greater sector dependence
0.50
0.40
0.30
0.20
HHI
1970
1980
1990
2000
2010
2020
Year